Drone Phenology Monitoring

From drone time-series to a map of vegetation phenology

Date: 08 Oct, 2026

Gaurav Rajput
Suryansh

Supervisor: Prof. Aaditeshwar Seth

Contents

1 What it does

2 How it works

3 The crown detector: Detectree2

4 Detection engineering

5 Crown tracking across time

6 Phenology analysis

7 Which orthomosaics work

8 The interactive viewer

9 What you get

10 Consensus crowns overlay

11 Trying different parameters

12 Walkthrough

What it does

1
Upload
A time-series of drone orthomosaics of your site.

2
Detect
Detectree2 outlines every tree crown in each image.

3
Track
Case-classified graph matching links the same crown across dates.

4
Classify
Phenology scores label each crown deciduous or evergreen.

5
Export
Interactive map, GeoJSON, CSV and GeoPackage files.

Everything is saved per run, so you can change the settings and try again whenever you like.

How it works

Six scripts, orchestrated by a bash runner. Two conda environments: **dpm-detectree** (GPU detection) and **dpm-tracking** (everything else).

STEP 0
Discover OMs
Scan folder, read GSD and CRS, sort chronologically, write pipeline_config.json.

STEP 1
Crown Detection
Multi-threshold Detectree2 on every OM. GPU accelerated.

STEP 2
Crown Tracking
PCC alignment, case-classified graph matching, consensus crown generation.

STEP 3
Phenology Analysis
Extract vegetation indices, compute deciduous scores, classify phenophase.

STEP 4a
COG Tiling
Convert orthomosaics to Cloud Optimized GeoTIFFs and XYZ tiles.

STEP 4b
Interactive Viewer
Leaflet map with timeline slider, crown filtering, phenology overlays.

The crown detector: Detectree2

What it is

Mask R-CNN instance segmentation, built on Meta's detectron2. Each tile returns crown polygons with confidence scores. Published weights used as-is — nothing is retrained.

Multi-threshold detection

Each OM is run at **multiple confidence thresholds** (base 0.45, alignment 0.65). This gives the tracker more candidates to match across dates.

Available model weights

Weight	Best for
flexi	Generalist: urban + closed canopy. Default.
urban_cambridge	Urban trees, Cambridge UK.
paracou	Tropical canopy, Paracou UAV.
tropical_closed	Dense tropical forest only.
randresize	Random-resize full model.

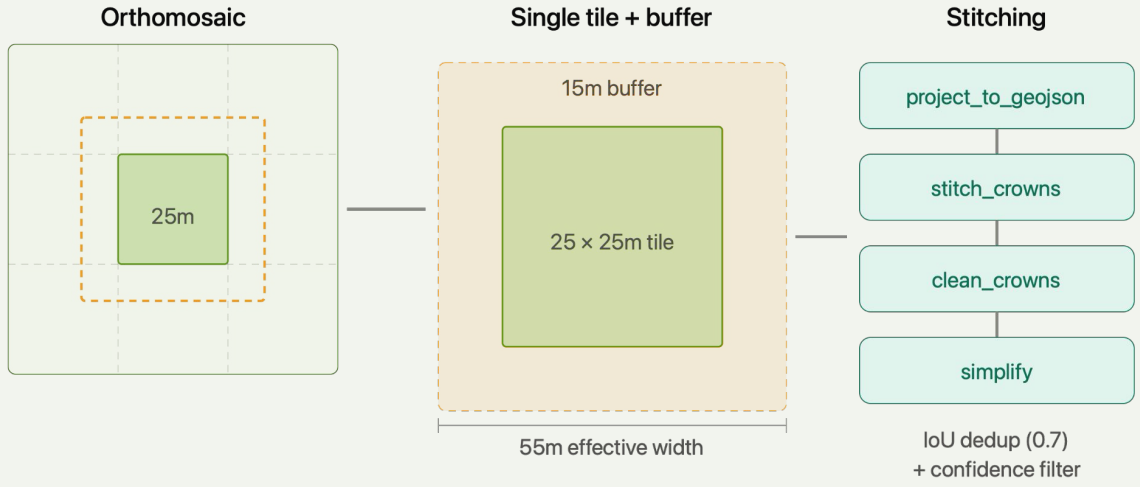
Mounted read-only at /models inside the container.

Detection engineering

Tiling

Each orthomosaic is used at its **native resolution.** The OM goes into Detectree2's tiler exactly as it is.

The ortho is cut into **25 × 25 m** tiles with a **15 m** buffer overlap, **all configurable by the user**. The buffer catches crowns that intersect a tile boundary.



Tiles back to one map

project_to_geojson Tile pixel masks become map coordinates.

stitch_crowns Per-tile GeoJSONs joined into one layer.

clean_crowns IoU-based dedup (fixed_iou=0.7) + confidence filtering, run per threshold layer in the multi-threshold GPKG.

simplify Geometry simplification at simplify_tol (default 1.0) to reduce file size.

Crown tracking across time

The same tree must be identified across every orthomosaic in the time-series. This is the core step that makes phenology monitoring possible.

1. Alignment

PCC-tiled phase cross-correlation estimates x/y shifts between consecutive OMs. Shifts logged for diagnostics.

2. Candidate pairs

For each crown pair within base_max_dist, compute IoU, overlap-prev, overlap-curr, centroid distance, area ratio, and shape similarity.

3. Case-classified edges

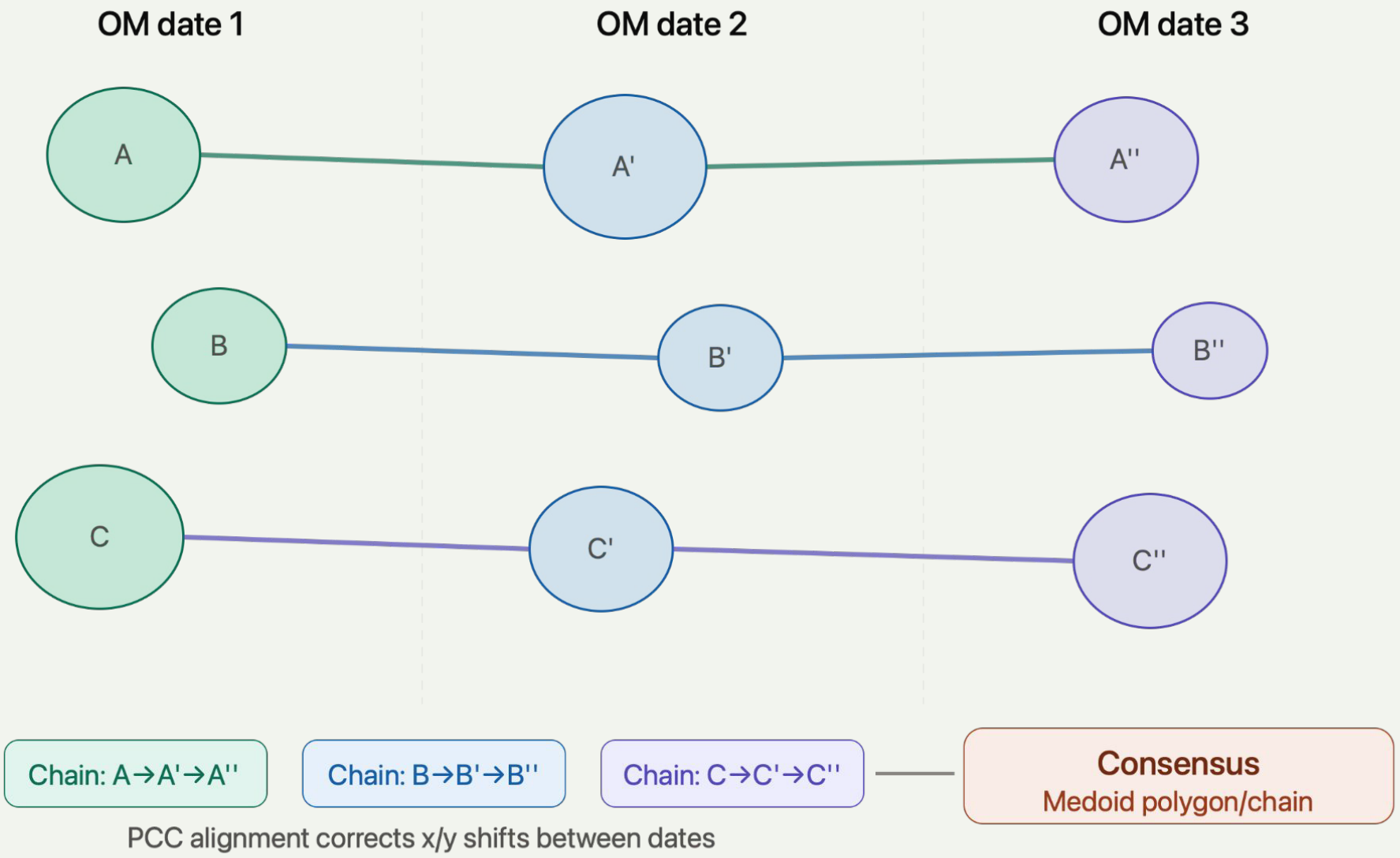
Each pair classified as **one_to_one**, **containment**, **nearby**, or **none**. Edges selected in that priority order into a NetworkX DiGraph.

4. Chain assembly

Greedy best-successor traversal by similarity score. Partial chains kept if they pass min length and one-to-one ratio ≥ 0.9.

Output: consensus_crowns_complete_all.gpkg — one consensus medoid polygon per chain.

Crown tracking across time

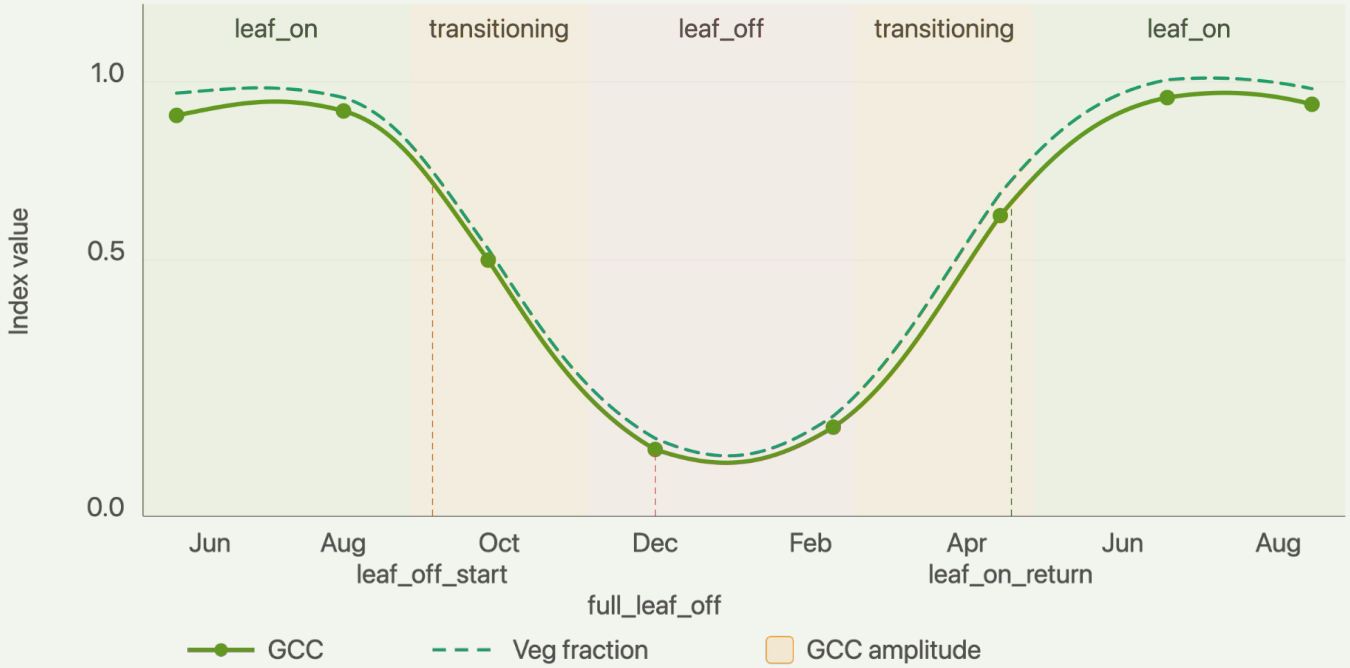


Phenology analysis

Vegetation features per crown

For each consensus crown across all orthomosaics, extract image-patch features: GCC (green chromatic coordinate), RCC (red chromatic coordinate), vegetation fraction via HSV mask, Shannon entropy of grayscale, and Laplacian variance. Laplacian variance also acts as patch QC — blurry patches are excluded.

A **deciduous score** (DS) is a weighted sum of three amplitude scores: veg-fraction amplitude (w=0.35), GCC amplitude (w=0.25), and texture amplitude via Laplacian variance. Threshold **DS ≥ -0.145** separates deciduous from evergreen.



Phenophase labels

Each deciduous (crown, OM) gets one of three labels:

leaf_on **transitioning** **leaf_off**

Leaf-shed events

Per deciduous chain: **leaf_off_start**, **full_leaf_off**, and **leaf_on_return** OM indices - the dates climate studies need.

Which orthomosaics work

What the app accepts

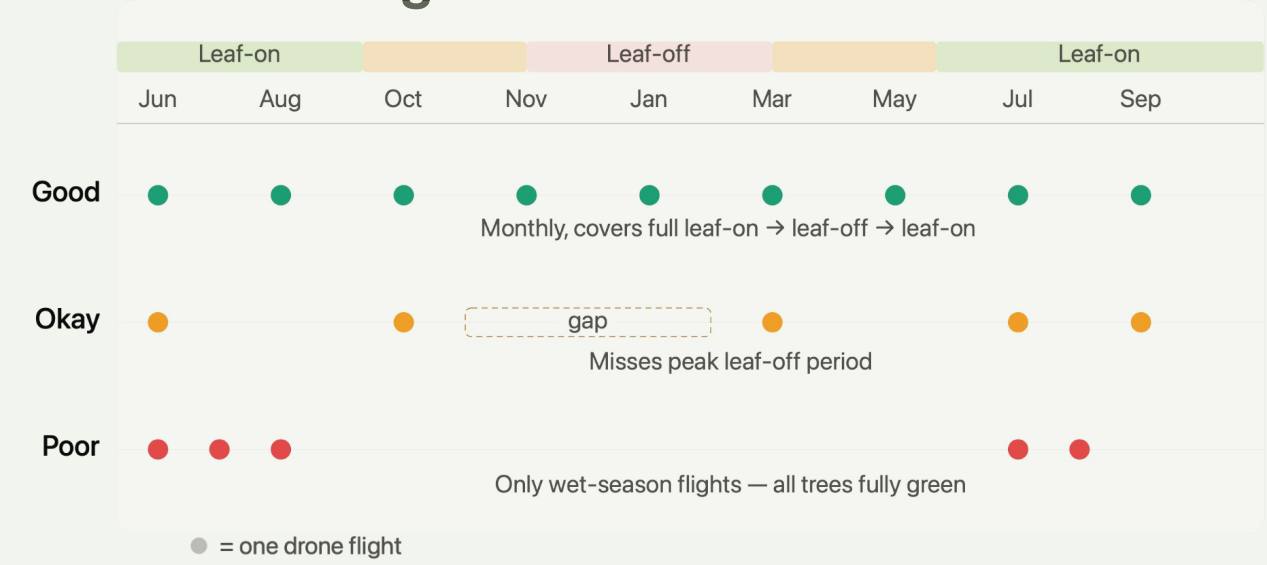
GeoTIFF (.tif or .tiff) with its map coordinates built in.

Best results with a ground resolution finer than 8 cm per pixel.

A **time-series** of at least 5 dates covering one full growing season is recommended.

Upload from your computer, or paste a Google Drive link.

What makes a good time-series



No EPSG? The app infers a likely UTM zone from the image coordinates and asks you to confirm — one click for Indian sites.

The interactive viewer

Step 4b generates a standalone HTML viewer (Leaflet-based) that opens in any browser. No server needed to view it.

Timeline slider — scrub through OM dates. Side-by-side comparison mode.

Crown comparison — two-slot panel with independent tile maps and sliders.

Phenology filtering — All / Deciduous / Evergreen.

Search — find crowns by ID or species annotation.

Crown info panel — click a crown to see GCC curve, deciduous score, leaf-off dates, per-OM tile strip.

Dark / light toggle — the viewer stays readable in sun-lit field conditions.

What you get

Interactive viewer

Standalone HTML map. Opens in any browser, no install needed.

GeoJSON

tree_master_geojson.geojson with crowns, scores, and phenophase labels.

CSV tables

Phenology features, leafshed scores, and phenophase-by-OM tables.

Crown GeoPackage

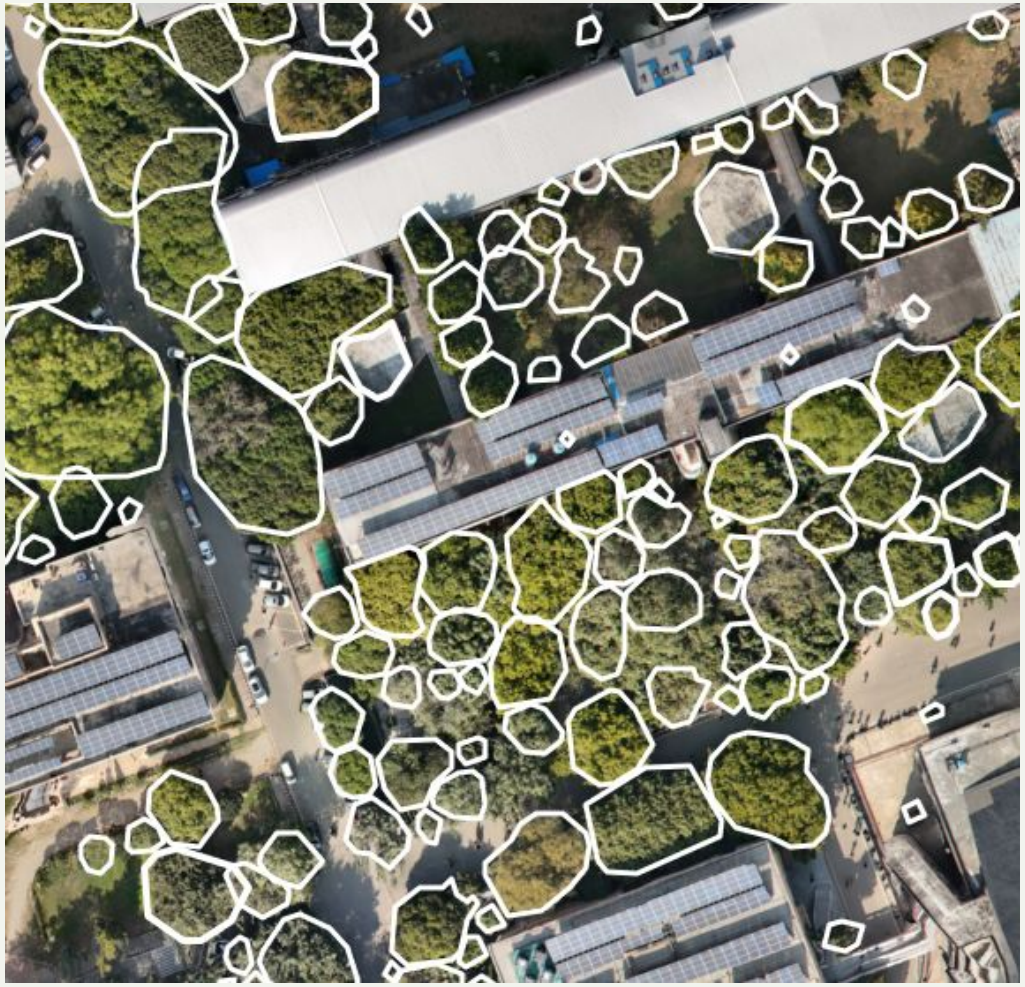
Consensus crown polygons with tracking metadata. Opens in QGIS.

Diagnostics

Alignment shifts, match rates, chain length distribution, tracking quality report.

Every run is kept. You can revisit any run later.

Consensus crowns overlay

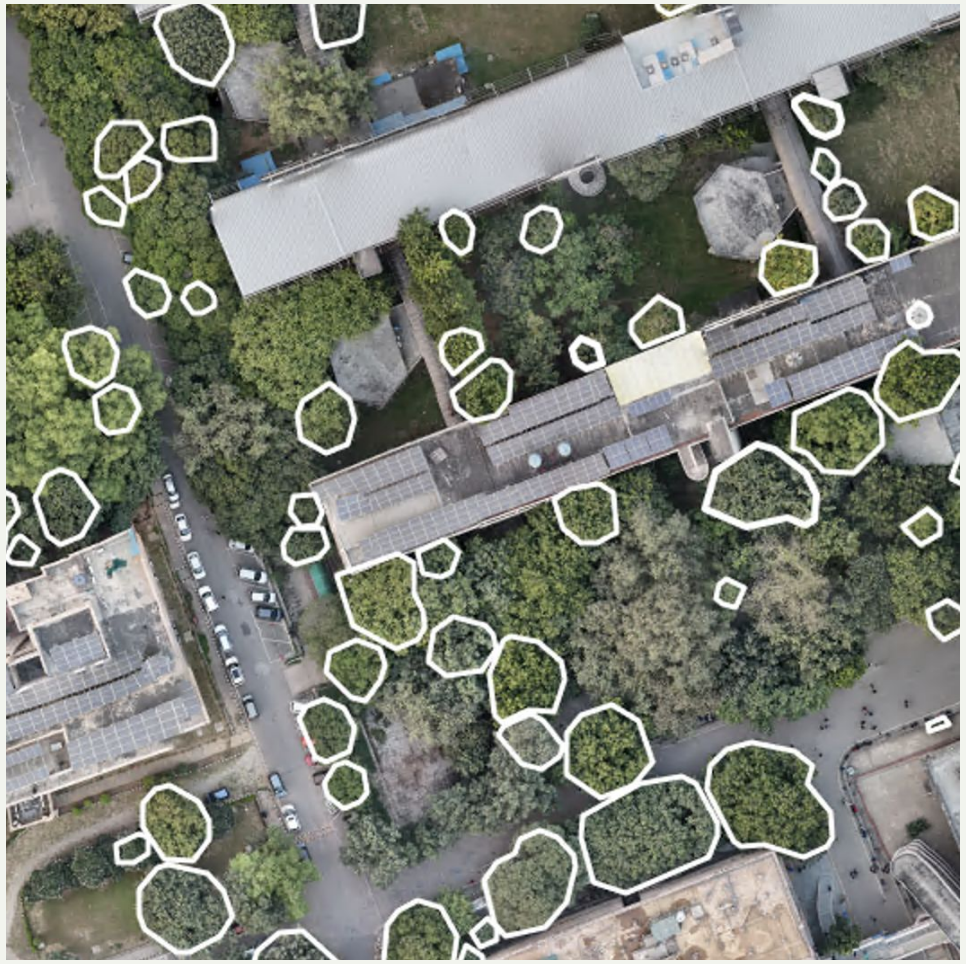


*19 orthomosaics were taken to generate these consensus crowns.

- For each tracked crown, the polygon most geometrically consistent with the rest by centroid distance, IoU, and area ratio is selected as the consensus crown.
- Trees with no outlines were missed.
- Objects outlined that are not tree crowns can be fixed by tuning parameters.
- On selecting a crown you can see the phenophase data for that crown.
- The interactive viz also has the deciduous and evergreen classification.

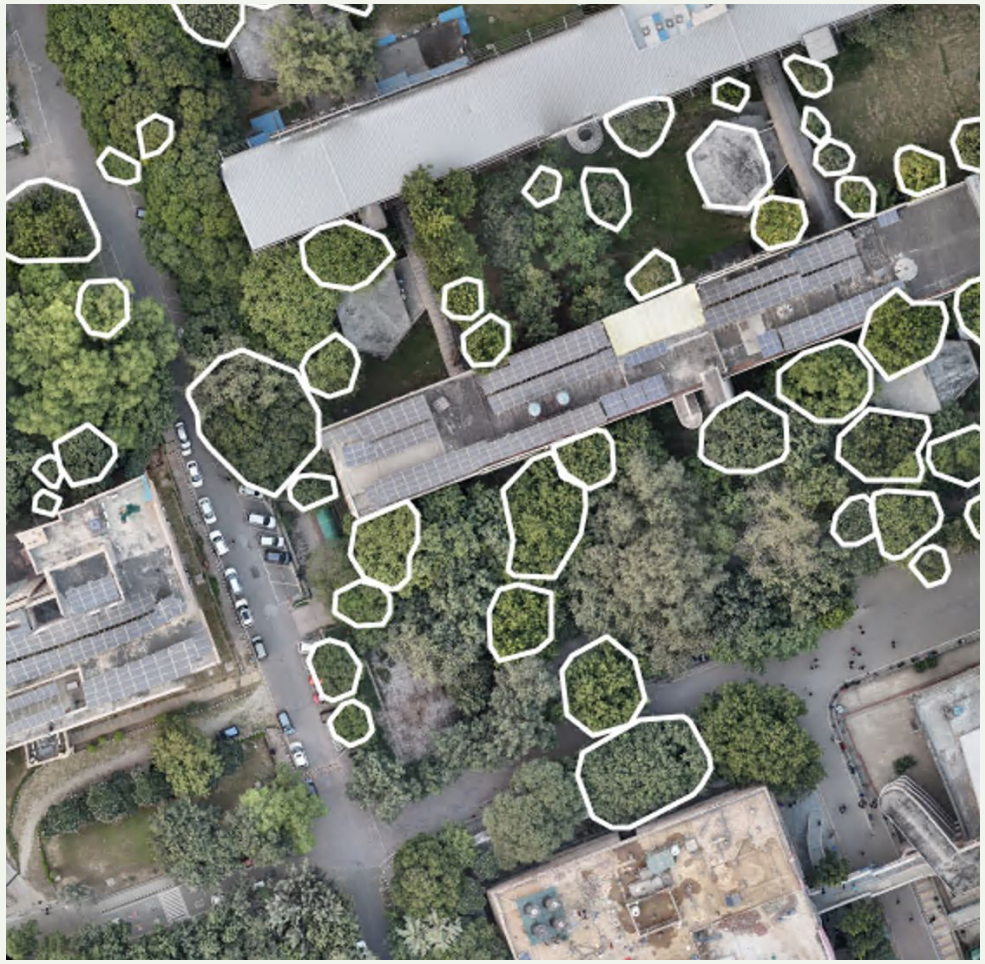
Trying different parameters

No. of orthomosaics- 5
Tile width - 25m
Tile height - 25m



Detected 94 crowns.

No. of orthomosaics - 5
Tile width - 40m
Tile height - 40m



Detected 75 crowns.

Thank you